CESFusion: Cross-Frequency Enhanced Spatial–Spectral Fusion Network for Hyperspectral and Multispectral Image Fusion

Haozheng Zhang, Yanhong Yang, Chaoyang Li, Yanjie Lu, Guodao Zhang, *Member, IEEE*, and Shengyong Chen, *Senior Member, IEEE*

***Abstract*— The fusion of hyperspectral and multispectral images (MSIs) involves integrating a high spectral resolution hyperspectral image (HSI) and a high spatial resolution MSI to generate an HSI with high-resolution HSI (HR-HSI). Existing HSI-MSI fusion methods primarily focus on information fusion within the spatial domain; however, few solutions have explored the employment of frequency analysis to enhance spatial resolution, limiting their capability for global perception. In this article, we propose an efficient and novel paradigm for HSI-MSI fusion through the cross-frequency enhanced spatial–spectral fusion network, named CESFusion, exploring the complementary fusion of information between the spatial and frequency domains. Specifically, we first present the cross-frequency fusion module (CFFM) to perform global analysis through the Fourier transform (FT) and effectively integrate and enhance the frequency domain information from both HSI and MSI. Subsequently, we propose the spectral modeling module (SpeMM) based on a state space model (SMM) to capture long-range spectral dependencies with linear complexity, and integrate it with the spatial residual block-based module (SRM) for joint spatial–spectral feature extraction. Finally, to enable sufficient interaction between the spatial and frequency domains, we adopt the cross-domain interaction module (CDIM), capturing and integrating complementary information from both domains. Moreover, a frequency-based loss function is purposely designed to further improve the restoration of global information. Extensive experiments conducted on both synthetic and real datasets demonstrate the superiority of our CESFusion, as evidenced by both quantitative and qualitative evaluation results.**

***Index Terms*— Frequency enhancement, hyperspectral and multispectral image (MSI) fusion, spatial–spectral learning, state space model (SMM).**

Received 25 February 2025; revised 26 March 2025; accepted 20 April 2025. Date of publication 23 April 2025; date of current version 8 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62406224, Grant 62020106004, and Grant 62102285; in part by the Natural Science Foundation of Tianjin under Grant 23JCQNJC00820 and Grant 22JCQNJC01100; and in part by the Fundamental Research Funds for the Provincial Universities of Zhejiang under Grant GK239909299001-019. *(Corresponding author: Yanhong Yang.)*

Haozheng Zhang and Yanhong Yang are with the School of Computer Science and Engineering and Tianjin Key Laboratory of Intelligence Computing and Novel Software Technology, Tianjin University of Technology, Tianjin 300384, China (e-mail: zhanghaozheng@stud.tjut.edu.cn; yyh@email.tjut.edu.cn).

Chaoyang Li and Shengyong Chen are with the Engineering Research Center of Learning-Based Intelligent System (Ministry of Education), the Key Laboratory of Computer Vision and Systems (Ministry of Education), and the School of Computer Science and Engineering, Tianjin University of Technology, Tianjin 300384, China (e-mail: lcy_822@stud.tjut.edu.cn; sy@ieee.org).

Yanjie Lu and Guodao Zhang are with the Institute of Intelligent Media Computing, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: yanjielu@hdu.edu.cn; guodaozhang@hdu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3563847

I. Introduction

HYPERSPECTRAL image (HSI) with hundreds of spectral narrow bands enables precise identification of different types of materials, ground objects, and crops [1]. The high spectral resolution enables a wide range of real-world and valuable applications, including classification [2], mineral exploitation [3], and intelligent agriculture [4]. On the other hand, multispectral image (MSI) offers high spatial resolution but exhibits significantly lower spectral resolution. Moreover, a single imaging platform cannot generate images with both high spatial and spectral resolution, importantly limiting further applications that require both [5]. A necessary compromise between spatial and spectral resolution is typically designed in image acquisition equipment. Fortunately, integrating high spectral resolution of HSI with high spatial resolution of MSI can effectively bridge the hardware limitation, resulting in high-resolution HSI (HR-HSI), commonly referred to as HSI-MSI Fusion.

Nowadays, numerous HSI-MSI fusion methods [6] have been developed to improve the spatial resolution of HSI, generally classified into decomposition-based approaches, pan-sharpening-based approaches, and deep learning (DL)-based approaches. Decomposition-based approaches mainly depend on different modeling techniques for HSI and MSI, including matrix factorization and tensor decomposition. Matrix factorization-based approaches [7] treat HSI and MSI as the product of the spectral basis matrix and coefficients matrix, leveraging the sparse and low-rank characteristics of both HSI and MSI [8], [9]. Tensor decomposition-based approaches primarily rely on tensor decomposition to represent HSI and MSI, such as Tucker decomposition [10], low-rank tensor factorization [11], and tensor ring decomposition [12]. Pan-sharpening-based methods fuse HSI and panchromatic image (PAN) to improve the spatial resolution of

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

HSI, often considered as special cases of HSI-MSI fusion. It is mainly subdivided into three categories: multiresolution analysis (MRA)-based approaches [13], component substitution (CS)-based approaches [14], and variational optimization-based approaches [15]. MRA-based approaches decompose images into different components at multiple resolutions and then reconstruct these components to enhance the resolution of the object image [16]. CS-based approaches separate HSI into spatial and spectral components, utilizing the spatial component of MSI to replace the corresponding component of HSI [17]. Variational optimization-based approaches construct an objective function to mathematically describe various constraints and regulations, and employ appropriate optimization methods to obtain the best solution, thus achieving HSI pan-sharpening [18].

DL-based approaches demonstrate impressive performance in HSI-MSI fusion tasks, due to their strong feature perceptual abilities [19]. As the powerful advantages of local information learning, convolution neural network (CNN)-based fusion methods have been extensively developed for spatial feature extraction in recent years [20], [21], [22]. For example, Liu et al. [23] proposed a fusion framework that integrates CNNs with pyramid split attention to enhance the spatial resolution of fused images. Xu et al. [24] presented a sophisticated architectural design, incorporating a network-in-network convolution block and batch normalization to improve feature extraction. Although existing CNN-based methods can effectively integrate the spatial–spectral features to some extent, they lack adaptive adjustments to local textural details based on long-range spatial feature maps, leading to spatial distortion in fused images. With strong long-distance dependencies modeling ability, Transformer-based methods have been applied to effectively extract global spatial features from input HSI and MSI [25], [26], [27]. Sun et al. [28] proposed a deep cross-fusion Transformer with multiscale mechanism to capture nonlocal information in images. Jia et al. [29] designed a two-branch Transformer network, integrating spectral features from HSI and spatial features from MSI through cross-modality concatenation. However, Transformer-based methods inevitably exhibit quadratic complexity due to their focus on nonlocal information, particularly when handling data with large-scale spatial resolutions and hundreds of channels. Recently, state space model (SSM), represented by Mamba [30], provides a novel architecture to address this problem, effectively utilizing linear complexity to achieve global information modeling. SSM-based methods have been extensively applied in various vision tasks [31], [32], effectively modeling long-distance spatial dependencies while requiring fewer computational resources than Transformer-based methods.

Despite the strong spatial modeling capabilities, most DL-based approaches mentioned above neglect the inter-spectral correlations and continuity of HSI, leading to spectral distortion in fused images. How to efficiently excavate spectral information from original HSI, minimize spectral distortion during fusion, and preserve the spectral quality of fused image remain important and challenging problems. Moreover, DL-based methods primarily concentrate on sophisticated network architectures to improve spectral and spatial feature learning in the spatial and spectral domains, while neglecting the importance of frequency domain information, which can more effectively characterize the global spatial properties of images.

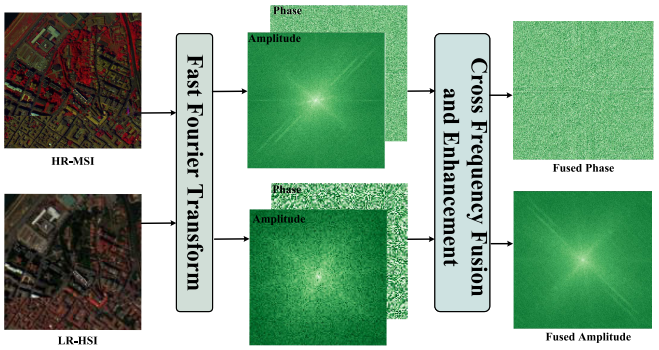


Fig. 1. Illustration of the amplitude and phase components extraction of HR-MSI and LR-HSI, and the results of the cross-frequency fusion of them.

To address the above-mentioned issues, we propose a novel cross-frequency enhanced spatial–spectral fusion framework, named CESFusion, effectively integrating frequency and spatial–spectral features to improve fusion performance. Specifically, the cross-frequency fusion module (CFFM) is designed to perform global analysis in the frequency domain. It leverages the Fourier transform (FT) to convert the input image into the frequency domain, decomposes it into amplitude and phase components, and then integrates the corresponding components of HSI and MSI to achieve cross-frequency interaction and enhance global perceptual effectiveness. Efficiently excavating spectral features is crucial for maintaining the spectral fidelity of fused images. We design the spectral modeling module (SpeMM) based on the SSM, in which the selective scanning mechanism enables explicit modeling of long-range spectral dependencies with linear computational complexity, further effectively preserving the spectral continuity of fused images. At the same time, a spatial residual block-based module (SRM) is developed to improve local feature extraction from HSI and MSI, and potentially decrease the loss of contextual features. Subsequently, the cross-domain interaction module (CDIM) is designed to facilitate complementary information interaction between spatial–spectral and frequency domain features. Moreover, a frequency-based loss function is also introduced to improve the model's ability to reconstruct global information. In summary, the major contributions of our article are as follows.

1) We introduce a novel approach for the HSI-MSI fusion, named CESFusion, to emphasize the significance of the frequency domain in enhancing global perceptual capabilities and facilitating cross-domain interaction and fusion of complementary information.
2) To improve general information perception, the CFFM is developed for frequency domain feature extraction,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

cross-frequency fusion, and enhancement. A frequency-based loss function is also specifically tailored to enhance the reconstruction of general information.

3) We propose the SpeMM to model the long-range spectral correlations and integrate it with the SRM to achieve joint spatial–spectral feature learning, effectively capturing rich spatial context while preserving spectral continuity. Extensive experiments conducted on both synthetic and real datasets demonstrate that our approach outperforms other methods in preserving spectral fidelity and enhancing spatial resolution.

II. Related Work

A. Traditional Fusion Methods

Traditional HSI-MSI fusion methods mainly leverage the intrinsic priors of HSI and MSI, such as low-rankness spatial characteristics and inter-spectral correlations, to construct fusion models, improving the spatial resolution of HSI. Most approaches use matrix factorization or tensor decomposition to model the fusion process, based on the aforementioned prior information. For example, Lanaras et al. [33] decomposed HSI and MSI into spectral reflectance and associated coefficients, performing super-resolution (SR) based on coupled matrix factorization. Yokoya et al. [34] proposed coupled nonnegative matrix factorization (CNMF) to alternately unmix HSI and MSI into endmember and abundance matrices. Equivalent to the nonlocal similarity of HR-HSI, Dian et al. [35] presented the low tensor-train rank (LTTR) prior and formulated an optimization problem under the LTTR constraint. Xue et al. [36] established the structured sparse low-rank representation of both spatial and spectral domains for HSI SR. Xu et al. [37] first constructed a nonlocal tensor at the patch level, and then adopted a unified objective function which fully incorporates spatial and spectral priors, well preserving both spatial and spectral similarity. Leveraging shift-invariant subspace properties, Xue et al. [38] introduced the tensor convolution-like low-rank dictionary to represent high-dimensional HSI. Unlike decomposition-based approaches, pan-sharpening-based approaches generally inject the high-resolution spatial components from MSI or PAN into HSI, adopting a compositional analysis perspective. Zhong et al. [14] incorporated the advantages of both CS and the MRA methods, significantly reducing spectral distortion while improving spatial quality. Based on the local gradient constraint, Fu et al. [15] designed a novel variational model and formulated the pan-sharpening objective as a convex optimization problem. However, traditional methods typically formulate a fusion objective function based on the intrinsic characteristics of the original images, neglecting the complex relationships and differences between HSI and MSI, which inevitably limit their generalizability.

B. DL-Based Fusion Methods

DL-based fusion methods have been extensively proposed to extract deep spatial and spectral features from HSI and MSI, capitalizing on their powerful feature learning capabilities [39]. CNN can effectively extract spatial features from images through local connections and weight-sharing mechanisms, which have been widely applied in HSI-MSI fusion. For instance, Liu et al. [40] proposed a two-branch CNN-based model to separately extract spatial and spectral features from both HSI and MSI, performing image fusion and restoration at the feature level. Zhang et al. [41] proposed a spatial–spectral reconstruction network (SSRNet), which decomposes HSI reconstruction into spatial and spectral components and incorporates spatial and spectral loss as constraints to improve interpretability. Considering the observational model of HR-HSI, Xie et al. [42] formulated fusion model parameters based on CNN and employed the proximal gradient approach to iteratively obtain the optimal solution. Ran et al. [43] proposed GuidedNet, utilizing a high-resolution guidance branch for multiscale decomposition and a feature reconstruction branch for detail reconstruction. To enhance model flexibility, Dong et al. [44] exploited both domain information likelihood and deep image prior, processed by a deep HSI denoiser, and developed a model-guided deep CNN (MoG-DCN) to unfold the HSI SR algorithm. In addition to deep spatial and spectral information, both the deep image prior and degradation model information also serve guiding roles in HSI-MSI fusion [45]. Li et al. [46] leveraged the degradation knowledge learning and a multiscale attention fusion to present an unsupervised model-guided coarse-to-fine fusion network for HSI SR. Liu et al. [47] further integrated spectral correlation, degradation knowledge, and structure prior with CNN to design an unfolding network with disentangled spatial–spectral representation for HSI SR. Despite the commendable performance of the CNN-based methods, there are distinct limitations in modeling long-range dependencies, which are essential for HSI-MSI fusion.

Recently, the explorations of Transformers in computer visions have also shown great potential for HSI-MSI fusion, as it can pay more attention to global relationships. Hu et al. [48] first adopted the Transformer-based architecture (Fusformer) to model the intrinsic dependencies within features, aiming to achieve HSI SR. To further improve the utilization of spatial information, Wang et al. [49] presented a multihierarchical cross Transformer (MCT-Net), efficiently integrating cross-modality information and reconstructing HR-HSI. Xiao et al. [50] introduced the Top-k Token Selective Transformer, employing a Residual Token Selective Group for compact self-attention and a Global Context Attention to focus on informative components. Deng et al. [51] combined the dilation spatial multihead self-attention and grouped spectral multihead self-attention to formulate a bidirectional dilation Transformer for HSI-MSI fusion. To enhance the cross-modality information interaction, Ma et al. [52] introduced a dual cross Transformer-based fusion method (DCTransformer) to capture the interplay and transfer between the two modalities. Chen et al. [53] designed cyclic interactive architecture based on Transformer (CYFormer) and introduced an adaptive interaction mechanism to fully exploit the spectral and spatial priors for enhancing fusion progress. Incorporating the deep unfolding method, Fang et al. [54] leveraged mutual information and deep priors to facilitate a degradation model-informed fusion method based on Transformer. Despite

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

the outstanding performance of Transformer-based approaches in long-range modeling, the quadratic complexity associated with high-resolution inputs is inevitable, leading to a significant computational burden.

C. State Space Models

Nowadays, SSMs have garnered significant interest for their ability to capture long-range dependencies while maintaining linear scalability [55]. Building on foundational SSM research [56], SSM-based solutions have been extensively explored in remote sensing image processing [57], [58]. Li et al. [59] introduced Mamba for HSI classification, adaptively modeling long-distance feature interaction and integration. Zhao et al. [60] combined CNN with Mamba to explore both global and local feature relationships, leveraging mutual guidance to enhance the resolution of fused images. Zhou et al. [61] designed a lightweight solution based on Mamba to achieve the global context learning for remote sensing satellite images dehazing. Peng et al. [62] expanded the single-input Mamba block into a dual-input FusionMamba block for effective spatial and spectral information integration. Xiao et al. [63] proposed a frequency-assisted Mamba framework to explore the spatial and frequency relationships in remote sensing image SR. Zhu et al. [64] integrated Mamba with implicit neural representation to produce MCIFNet, which effectively generates high-quality fused images in a continuous domain through point-to-point processing. However, there are few research to employ the great potential of Mamba in exploring effective spatial and spectral learning, which is necessary in HSI-MSI fusion.

D. FT in Computer Vision

The FT has been fundamental in frequency domain understanding, easily enabling the analysis of global statistical information. This technique has been extensively explored in computer vision tasks [65], [66], [67]. In terms of image fusion, Zhou et al. [68] proposed a general spatial-frequency integration network (SFIN) to capture the frequency information, and achieved the dual-domain representation interaction in pan-sharpening and depth SR tasks. Zheng et al. [69] devised the frequency integration and spatial compensation network (FISCNet) to delve into both spatial and frequency domains simultaneously in the infrared and visible image fusion, maintaining the visual perception and improving the detailed texture. Taking advantages of the FT in global modeling, Yang et al. [70] exploited the amplitude and phase components of the frequency domain to facilitate the multiexposure images' interaction and fusion. Despite these advancements, the application of the FT in HSI-MSI fusion remains underexplored, suggesting significant potential for performance enhancement by adopting FT-based fusion architectures.

III. Preliminaries

A. State Space Models

SSMs are fundamental for mapping 1-D sequence $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through an intermediate state $h(t) \in \mathbb{R}^N$, motivated by classical continuous systems. This workflow can be represented as a linear ordinary differential equation (ODE)

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ denotes the evolution parameter, while $\mathbf{B} \in \mathbb{R}^{N\times 1}$ and $\mathbf{C} \in \mathbb{R}^{1\times N}$ represent the projection parameters along the depth dimension.

To better adapt to DL applications, a discretization process utilizing the zero-order hold (ZOH) methodology is incorporated to enable adaptive scanning and adjustment of the input data. After discretization, the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ are transformed into their discrete parameters $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$, respectively. Then, linear recurrence or global convolution can be exploited to compute the SSM-based models, as shown in the following formulas:

$$\begin{aligned} h'(t) &= \overline{\mathbf{A}}h(t) + \overline{\mathbf{B}}x(t) \\ y(t) &= \mathbf{C}h(t). \end{aligned} \tag{2}$$

B. FT for Frequency Analysis

The FT serves as a mathematical technique that transforms input data from the time or spatial domain into its frequency representation. Specifically, given a single-channel image x with size of $H \times W$, the FT $\mathcal{F}$ can be formulated as follows:

$$\mathcal{F}(x)(u,v) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} x(h,w)e^{-j2\pi\left(\frac{h}{H}u+\frac{w}{w}v\right)} \tag{3}$$

where (u, v) represents the coordinate in Fourier space. This transform converts spatial features of the image into its frequency components. Moreover, these frequency components can be decomposed into a real part and an imaginary part, forming the amplitude component $\mathcal{A}$ and the phase component $\mathcal{P}$, which can be expressed as follows:

$$\begin{aligned} \mathcal{A}(\mathcal{F}(x)(u,v)) &= \sqrt{R^2(\mathcal{F}(x)(u,v)) + I^2(\mathcal{F}(x)(u,v))} \\ \mathcal{P}(\mathcal{F}(x)(u,v)) &= \arctan\left[\frac{I(\mathcal{F}(x)(u,v))}{R(\mathcal{F}(x)(u,v))}\right] \end{aligned} \tag{4}$$

where R and I denote the real and imaginary parts of the frequency representation $\mathcal{F}(x)(u, v)$, respectively, representing the structure information of the image.

IV. Method

A. Overview

Fig. 2 illustrates the overall architecture of the proposed CESFusion, including the SSM-based SpeMM for spectral feature learning, the SRM for local detailed feature extraction, the CFFM for frequency domain integration and enhancement between HSI and MSI, and the CDIM for complementary fusion of spatial–spectral and frequency domain features.

Given the low-resolution HSI (LR-HSI) $\mathbf{Z} \in \mathbb{R}^{h\times w\times C}$ and the high-resolution MSI (HR-MSI) $\mathbf{Y} \in \mathbb{R}^{H\times W\times c}$, we first apply bicubic interpolation to upsample $\mathbf{Z}$ to $\mathbf{Z}\uparrow \in \mathbb{R}^{H\times W\times C}$, and then adopt two embedding layers, consisting of 3×3 and 1×1 convolution layers, to extract shallow features from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

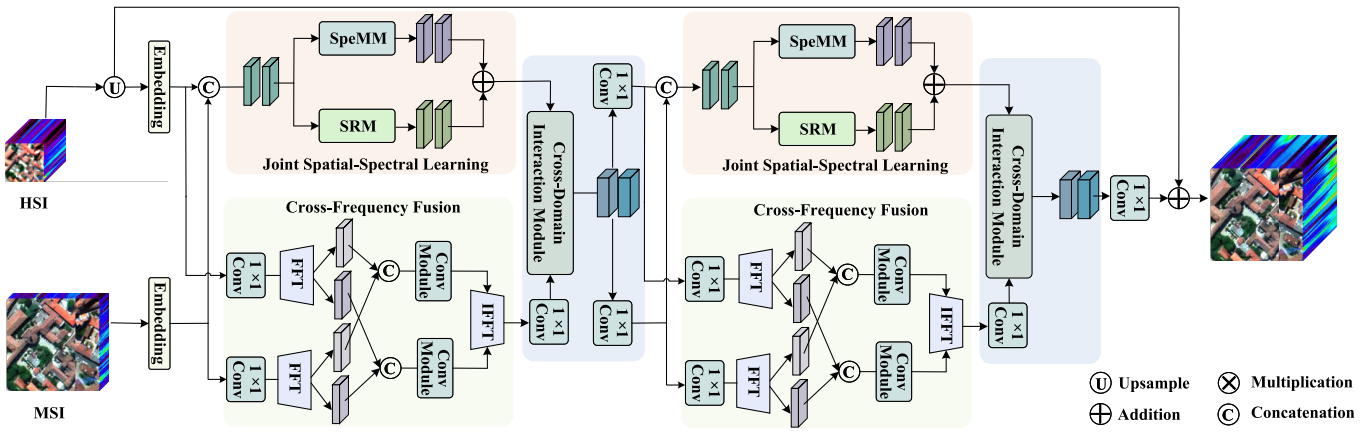


Fig. 2. Detailed framework of the proposed central component in the CESFusion, consisting of four functional modules: the SSM-based SpeMM, the SRM, the CFFM, and the CDIM.

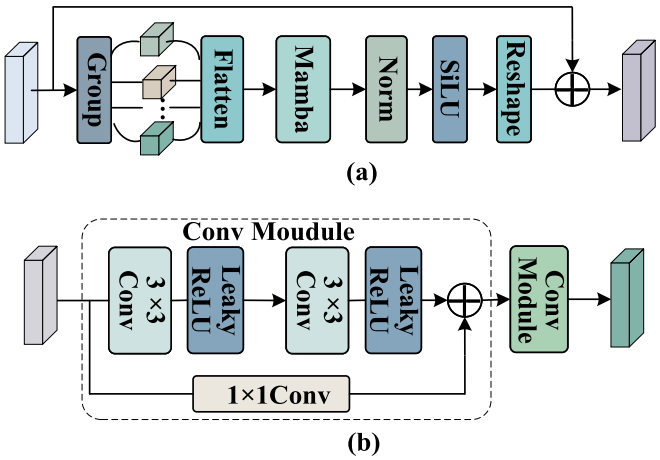


Fig. 3. (a) and (b) are illustrations of the SpeMM and SRM, respectively.

$\mathbf{Z}\uparrow$ and $\mathbf{Y}$, respectively. Subsequently, deep feature learning is carried out in two stages: first, joint spatial–spectral feature extraction and cross-frequency fusion are performed employing the SRM, SpeMM, and CFFM; second, the CDIM is applied for cross-domain feature fusion to generate the fused representation. To further enhance complementary fusion, two rounds of deep feature extraction are implemented. Finally, a 3×3 convolutional layer with the skip connection is adopted to restore the final HR-HSI.

B. Joint Spatial–Spectral Modeling

To achieve effective spatial–spectral integration, we directly concatenate the shallow embedded representations of HSI and MSI at the channel dimension. Then, we exploit the SRM and SpeMM to generate the joint spatial–spectral features. As depicted in Fig. 3, the SRM consists of residual blocks with 3×3 convolutional layers and PReLU activation layers, extracting local spatial features in the spatial domain. With residual connection, these blocks preserve more detailed information for complementary fusion.

The aforementioned SSM enables adaptive modeling of global features with linear computational complexity, utilizing the selective scanning mechanism. Therefore, we adopt the SpeMM to effectively model the dependencies among spectral bands based on SSM, maintaining the spectral continuity of fused HR-HSI. To reduce the computational complexity introduced by the large number of channels, we first divide the input features into T tokens along the channel dimension. These spectral tokens are then fed into the Mamba block to extract spectral features by capturing the inter-spectral correlations. Specifically, we extract the spectral features as follows:

$$\begin{aligned} X_1 &= \text{Flatten}(\text{Divide}(X)) \\ X_2 &= \text{SiLU}(\text{GN}(\text{Mamba}(X_1))) \\ X_3 &= \text{Reshape}(X_2) + X \end{aligned} \tag{5}$$

where $X, X_3 \in \mathbb{R}^{B\times H\times W\times D}$ and $X_1, \mathrm{X}_2 \in \mathbb{R}^{(B\times H\times W)\times T\times S}$ represent the input feature, the output feature, the flatten ones, and the features processed by the Mamba block, respectively. B, H, W, and D denote the batch size, height, width, and embedding dimension of the input feature, respectively. T is the number of divided tokens, and S is the number of channels in each token. Mamba$(\cdot)$ refers to the standard Mamba block presented in [30]. GN$(\cdot)$ is the group normalization operation. SiLU$(\cdot)$ is the Sigmoid linear unit, serving as an activation function.

C. Cross-Frequency Fusion

In the CFFM, as shown in Fig. 1, for the input features F_0^h and F_0^m, we first adopt a 1×1 covolutional layer to generate F_1^h and F_1^m, respectively. Then, we apply the fast FT (FFT) to obtain their frequency representations, specifically the amplitude and phase components. The detailed process is as follows:

$$\begin{aligned} \mathcal{A}^h, \mathcal{P}^h &= \mathcal{F}\left(\text{Conv}\left(F_1^h\right)\right) \\ \mathcal{A}^m, \mathcal{P}^m &= \mathcal{F}\left(\text{Conv}\left(F_1^m\right)\right) \end{aligned} \tag{6}$$

where $\mathcal{A}$ and $\mathcal{P}$ denote the amplitude and phase components, respectively. $\mathcal{F}$ is the FFT.

To integrate the complementary frequency information from HSI and MSI, we concatenate the corresponding amplitude

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

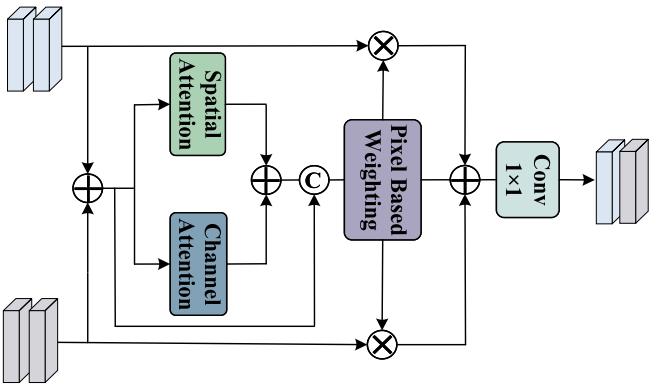


Fig. 4. Architecture of CDIM.

and phase components of both HSI and MSI, enabling cross-frequency domain fusion. Next, we adopt a ConvModule, consisting of a 1×1 convolution layer and the rectified linear unit (ReLU) activation layer, to enhance the frequency information representation, further improving the global perceptual ability. The details can be expressed as follows:

$$\begin{aligned} \mathcal{A}^f &= \mathrm{ReLU}\big(\mathrm{Conv}\big(\mathrm{Concat}\big(\big[\mathcal{A}^h, \mathcal{A}^m\big]\big)\big)\big) \\ \mathcal{P}^f &= \mathrm{ReLU}\big(\mathrm{Conv}\big(\mathrm{Concat}\big(\big[\mathcal{P}^h, \mathcal{P}^m\big]\big)\big)\big) \end{aligned} \tag{7}$$

where $\mathrm{Concat}(\cdot)$ denotes the channel-wise concatenation, and $\mathcal{A}^f$ and $\mathcal{P}^f$ represent the fused amplitude and phase components, respectively. At the end, we apply the inverse FFT (IFFT) to transform the global frequency representations back into the spatial domain features. This process can be expressed as follows:

$$F^{\mathrm{fre}} = \mathrm{Conv}(\mathcal{F}^{-1}(\mathcal{A}^f, \mathcal{P}^f)) \tag{8}$$

where $\mathcal{F}^{-1}$ is the IFFT operation, and F^{fre} is the output cross-fused frequency feature.

D. Cross-Domain Feature Interaction and Fusion

To adaptively integrate the frequency feature and the joint spatial–spectral feature, as illustrated in Fig. 4, the architecture of CDIM is designed as follows. We first combine the joint spatial–spectral feature F^{ss} with the frequency feature F^{fre} through an addition operation to produce the coarse fused feature F^{fus}. Then, we generate both the channel attention and spatial attention weights for the coarse feature simultaneously. These weights are then fused together through an addition operation to obtain the coarse attention weight W_{fus}. The detailed process can be formulated as follows:

$$\begin{aligned} F^{\mathrm{fus}} &= F^{\mathrm{ss}} + F^{\mathrm{fre}} \\ W_{\mathrm{spa}} &= \mathrm{Conv}(\mathrm{Cat}\big[\mathrm{GAP}(F^{\mathrm{fus}}), \mathrm{GMP}(F^{\mathrm{fus}})\big]) \\ W_{\mathrm{spe}} &= \mathrm{Conv}(\mathrm{GAP}(F^{\mathrm{fus}})) \\ W_{\mathrm{fus}} &= W_{\mathrm{spa}} + W_{\mathrm{spe}} \end{aligned} \tag{9}$$

where $\mathrm{GAP}(\cdot)$ and $\mathrm{GMP}(\cdot)$ denote the general average pooling and global max pooling operations, respectively. W_{spa} and W_{spe} are the spatial attention and spectral attention weights, respectively. To refine the coarse attention weight W_{fus}, we introduce a channel shuffle operation to incorporate the input feature with W_{fus}, producing the final fused weights W_{final}. This process employs the input feature as guidance to adjust W_{fus}. The process can be defined as follows:

$$W_{\mathrm{final}} = \mathrm{Sigmoid}(\mathrm{Conv}(\mathrm{CS}(W_{\mathrm{fus}}, F_{\mathrm{fus}}))) \tag{10}$$

where $\mathrm{CS}(\cdot)$ is the channel shuffle operation, and $\mathrm{Sigmoid}(\cdot)$ is applied to compute the final feature weight. Finally, we utilize adaptive weighted summation to obtain the final cross-domain fused feature

$$F_{\mathrm{final}} = \mathrm{Conv}(F^{\mathrm{ss}} \cdot W_{\mathrm{final}} + F^{\mathrm{fre}} \cdot (1 - W_{\mathrm{final}}) + F^{\mathrm{ss}} + F^{\mathrm{fre}}) \tag{11}$$

where a skip connection is also introduced to mitigate the gradient vanishing.

E. Loss Function

In addition to novel modules, we also formulate a novel loss function to further optimize the training process to enhance the quality of fused images in both spatial and frequency domains. It consists of two parts: a spatial-based loss and a frequency-based loss. In the spatial domain, we employ $L1$ loss to supervise the difference between the final output and the ground truth (GT)

$$L_{\mathrm{spa}} = \left\| X_{\mathrm{out}} - X_{\mathrm{gt}} \right\|_1. \tag{12}$$

Similarly, we also adopt $L1$ loss to the amplitude component and phase component, respectively, to formulate the frequency-based loss L_{fre}

$$\begin{aligned} L_{\mathrm{fre}} &= L_{\mathrm{amp}} + L_{\mathrm{pha}} \\ &= \left\| \mathcal{A}_{\mathrm{out}} - \mathcal{A}_{\mathrm{gt}} \right\|_1 + \left\| \mathcal{P}_{\mathrm{out}} - \mathcal{P}_{\mathrm{gt}} \right\|_1 \end{aligned} \tag{13}$$

where $\mathcal{A}$ and $\mathcal{P}$ denote the amplitude and phase component, respectively, which are obtained by FFT. Therefore, the final loss function can be described as follows:

$$L = L_{\mathrm{spa}} + \lambda L_{\mathrm{fre}} \tag{14}$$

where λ is the balance parameter between L_{spa} and L_{fre}. λ is empirically set to 0.7 in our implementation.

V. Experiments

This section is organized to present the experimental results that demonstrate the efficacy of our proposed CESFusion for HSI-MSI fusion. First, we provide details of the datasets. Next, we describe the experiment settings, including the data simulation process and implementation details. Following this, we outline the quality metrics adopted. Subsequently, we compare the fusion performance of CESFusion with state-of-the-art HSI-MSI fusion methods. Finally, we present the results of the corresponding ablation experiments and perform a complexity analysis.

A. Data Description

To comprehensively evaluate the performance of our CESFusion, we conduct experiments on four publicly available datasets, including Pavia Center [71], Botswana [72], Chikusei [73], and XiongAn [74]. The details of each dataset are presented below.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

1) Pavia Center: The Pavia Center dataset was acquired in 2001 using the ROSIS sensors. Initially, it consisted of 115 spectral bands covering from 430 to 960 nm; however, due to water vapor absorption, only 102 bands are available for analysis. The dataset has a spatial size of 1096 × 1096 pixels.

2) Botswana: The Botswana dataset was captured by the Hyperion sensor and originally contained 242 spectral bands ranging from 400 to 2500 nm. However, 97 bands were eliminated due to noise and calibration issues, leaving 145 bands available for the experiment. The dataset has a spatial size of 1496 × 256 pixels.

3) Chikusei: The Chikusei dataset consists of images taken by the Headwall Hyperspec-VNIR-C imaging sensor in Chikusei, Japan. The dataset originally consisted of 128 spectral bands, spanning from 363 to 1018 nm, and has a total spatial size of 2304 × 2304 pixels.

4) XiongAn: The XiongAn dataset was captured in XiongAn New Area (Matiwan Village), China. It has a spatial resolution of 1580 × 3750 pixels. It contains 256 spectral bands, ranging from 400 to 1000 nm. Compared to other datasets, it has more spectral bands, which further increases the difficulty of image fusion.

B. Experiment Setting

1) Data Simulation and Collection: In these experiments, we utilize LR-HSI and HR-MSI pairs as input, with HR-HSI serving as the GT. Therefore, we design a simulation process based on Wald's protocol [75] to generate LR-HSI and HR-MSI. Specifically, taking the Pavia Center dataset as an example, we apply a 8 × 8 Gaussian filter on each of the original images, serving as the HR-HSI. Then, we downsample the blurred images by a scaling factor of 4 to obtain LR-HSI. Subsequently, HR-MSI is generated by extracting five bands from the original images at equal intervals. We randomly crop six images with the spatial size of 32 × 32 as the test set, with the remaining serving as the training set, of which the spatial size is 16 × 16.

2) Implementation Details: To ensure a fair comparison, all DL-based methods are trained on a Linux operating system using Python 3.8.1 and PyTorch 1.13.0. The training process is carried out on an NVIDIA GPU GeForce GTX 3090, and the Adam optimizer is used to minimize the training loss. The weight decay is set to $1e^{-4}$, with an initial learning rate of 0.0001. The learning rate is reduced by a factor of 0.2 every 1500 epochs, and the total number of epochs is set to 5000.

C. Quality Metrics

In this study, six quality metrics are employed to evaluate the quality of fused images, including the peak signal-to-noise ratio (PSNR), the Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS), the spectral angle mapper (SAM), the correlation coefficient (CC), the root mean square error (RMSE), and the structural similarity (SSIM).

1) PSNR: PSNR is a commonly applied quality metric for quantifying the reconstruction quality of images. PSNR of the kth band is calculated as follows:

$$\mathrm{PSNR} = 10\lg\left(\frac{\max(X_k)^2}{\frac{1}{HW}\left\|X_k - \hat{X}_k\right\|_2^2}\right) \tag{15}$$

where X_k and $\hat{X}_k$ denote the reference HR-HSI and the generated HR-HSI at the kth band, respectively. H and W represent the height and width of the images. The function $\max(\cdot)$ returns the maximum value, and $\|\cdot\|_2$ denotes Euclidean (two-norm) function.

2) ERGAS: Considering the average brightness and contrast characteristics of the image, ERGAS is exploited to evaluate the overall error of the fused image

$$\mathrm{ERGAS} = \frac{100}{\mathrm{sf}}\sqrt{\frac{1}{L}\sum_{k=1}^{L}\frac{\left\|X_k - \hat{X}_k\right\|_2^2}{\mu^2(X_k)}} \tag{16}$$

where sf denotes the scaling factor, and the function $\mu(\cdot)$ calculates the mean value.

3) SAM: SAM is utilized to measure the similarity of spectral features between the fused image and the reference image

$$\mathrm{SAM} = \arccos\left(\frac{\langle X(i,j), \hat{X}(i,j)\rangle}{\|X(i,j)\|_2\|\hat{X}(i,j)\|_2}\right) \tag{17}$$

where $X(i,j)$ and $\hat{X}(i,j)$ represent the spectral vectors of the reference HR-HSI and the generated HR-HSI at the specified pixel position (i,j), respectively. Additionally, $\langle\cdot,\cdot\rangle$ denotes the inner product. The final SAM value is obtained by averaging the SAM values across all pixels.

4) CC: CC evaluates the similarity between the generated images and the reference images via calculating the CC between their pixel values for each band

$$\mathrm{CC} = \frac{1}{L}\sum_{k=1}^{L}\mathrm{corrcoef}\left(X_k, \hat{X}_k\right) \tag{18}$$

where corrcoef$(\cdot)$ represents the function that calculates the CC for each band, and L is the total number of bands in the image.

5) RMSE: RMSE can be employed to evaluate the similarity by calculating the pixel-wise differences between the reference and generated images

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{k=1}^{L}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(X_k(i,j) - \hat{X}_k(i,j)\right)^2}{LHW}} \tag{19}$$

where L, H, and W represent the number of bands, height, and width of the image, respectively.

6) SSIM: SSIM is applied to evaluate the similarity between two images by considering their correlation, contrast, distortion, and brightness

$$\mathrm{SSIM} = \frac{(2\mu_x\mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{\left(\mu_x^2 + \mu_{\hat{x}}^2 + C_1\right)\left(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2\right)} \tag{20}$$

where C_1 and C_2 are all constants, μ_x and $\mu_{\hat{x}}$ are the mean of X and $\hat{X}$, respectively. σ_x and $\sigma_{\hat{x}}$ denote the standard deviations of X and $\hat{X}$, respectively. $\mu_x^2 + \mu_{\hat{x}}^2$ represents the covariance of X and $\hat{X}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

TABLE I

AVERAGE QUANTITATIVE COMPARISON OF VARIOUS METHODS ON THE PAVIA CENTER AND THE BOTSWANA DATASETS. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLDFACE, AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	Pavia Center						Botswana					
	PSNR↑	ERGAS↓	SAM↓	CC↑	RMSE↓	SSIM↑	PSNR↑	ERGAS↓	SAM↓	CC↑	RMSE↓	SSIM↑
CNMF [34]	28.9133	7.3542	5.9812	0.9326	0.0570	0.7054	28.8083	2.4940	2.4603	0.9694	0.0252	0.8672
LTTR [35]	30.3303	5.4714	9.3015	0.9595	0.0805	0.7629	24.0969	16.0514	3.9607	0.8735	0.0784	0.6875
TFNet [40]	40.4911	1.7887	3.9612	0.9931	0.0048	0.9930	32.9766	1.4756	2.0932	0.9768	0.0085	0.9761
ResTFNet [40]	41.9797	1.6460	3.7215	0.9939	0.0022	0.9938	32.7811	1.2525	1.9136	0.9745	0.0062	0.9740
SSR-Net [41]	43.1656	1.5027	3.6627	0.9951	0.0011	0.9950	32.2983	1.3444	2.1239	0.9702	0.0026	0.8742
MoG-DCN [44]	37.0255	2.3193	4.8165	0.9893	0.0079	0.9890	28.9669	2.0942	2.7325	0.9572	0.0128	0.9555
Fusformer [48]	44.1733	1.4266	**3.4157**	0.9954	0.0018	0.9954	36.3853	0.9521	1.3739	0.9896	0.0084	0.9892
MCT-Net [49]	41.0756	1.7119	4.0242	0.9944	0.0025	0.9941	30.9513	1.5466	2.2911	0.9613	0.0075	0.9604
SFIN [68]	43.0659	1.5111	3.5855	0.9950	0.0021	0.9949	34.6729	1.0433	1.5805	0.9832	0.0033	0.9830
DCTransformer [52]	41.4938	1.6600	3.8719	0.9942	0.0028	0.9940	31.1946	1.3402	2.0715	0.9600	0.0073	0.9574
CYFormer [53]	42.4643	1.5547	3.7322	0.9948	0.0015	0.9948	33.9575	1.0560	1.6368	0.9751	0.0064	0.9747
FusionMamba [62]	39.0273	1.9639	4.2716	0.9922	0.0041	0.9920	31.7846	1.6289	2.3057	0.9713	0.0081	0.9710
MCIFNet [64]	42.7614	1.5392	3.6655	0.9952	0.0017	0.9949	35.0067	1.1302	1.6617	0.9845	0.0025	0.9839
CESFusion	**44.4736**	**1.4176**	3.4290	**0.9955**	**0.0011**	**0.9954**	**37.7721**	**0.8796**	**1.3132**	**0.9925**	**0.0023**	**0.9925**

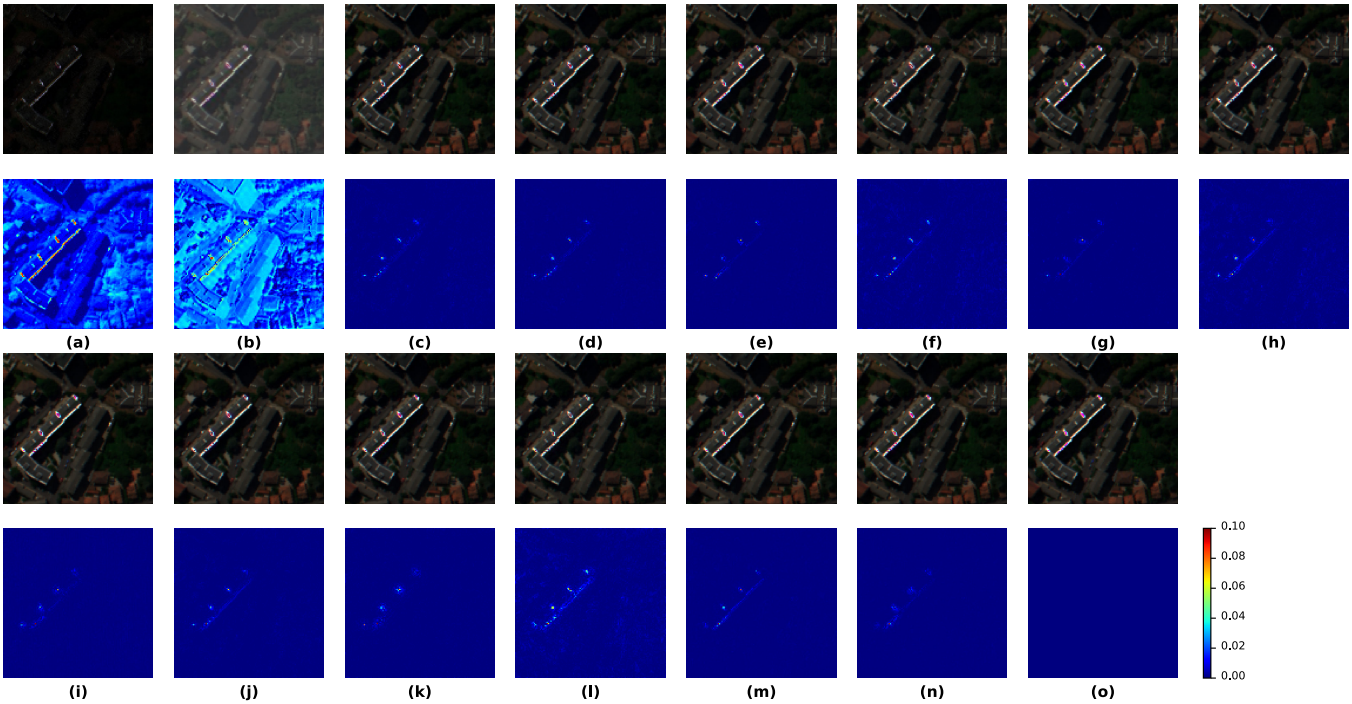


Fig. 5. Qualitative results of the Pavia Center dataset with pseudo-color technique. The first row presents fused images with bands 60-30-10 as RGB. The second row depicts the corresponding error images at the band of 75. (a) CNMF. (b) LTTR. (c) TFNet. (d) ResTFNet. (e) SSR-Net. (f) MoG-DCN. (g) Fusformer. (h) MCT-Net. (i) SFIN. (j) DCTransformer. (k) CYFormer. (l) FusionMamba. (m) MCIFNet. (n) CESFusion. (o) GT.

D. Performance Comparison

To demonstrate the superior performance of our proposed CESFusion, we conduct a series of comparative experiments with state-of-the-art algorithms on four openly accessible datasets. The compared methods include traditional methods such as CNMF [34] and LTTR [35]; CNN-based methods including TFNet [40], ResTFNet [40], SSR-Net [41], MoG-DCN [44], and SFIN [68]; Transformer-based methods such as Fusformer [48], MCT-Net [49], DCTransformer [52], and CYFormer [53]; and Mamba-based methods such as FusionMamba [62] and MCIFNet [64].

1) Results on the Pavia Center: The experimental quantitative results on the Pavia Center dataset are presented in Table I. It can be clearly observed that both traditional approaches, CNMF and LTTR, fail to provide superior results as DL-based approaches. Among all DL-based approaches, CESFusion achieves optimal performance across five metrics. It demonstrates obvious advantages in terms of global perceptual ability and joint spatial–spectral feature learning, effectively ensuring both spatial and spectral fidelity in the fused results. Fusformer achieves suboptimal fusion performance, based on the powerful capability of the Transformer in capturing long-distance dependencies. To provide a more intuitive fusion result, we utilize the pseudo-color technique to display both fused images and the error images, representing the difference between the reference HR-HSI and the output HR-HSI. As shown in Fig. 5, CESFusion preserves rich textural details, resulting in excellent spatial fidelity. Notably, significant block artifacts exist in the error images produced by traditional methods, indicating that simple prior information cannot guide the model to achieve optimal fusion results.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

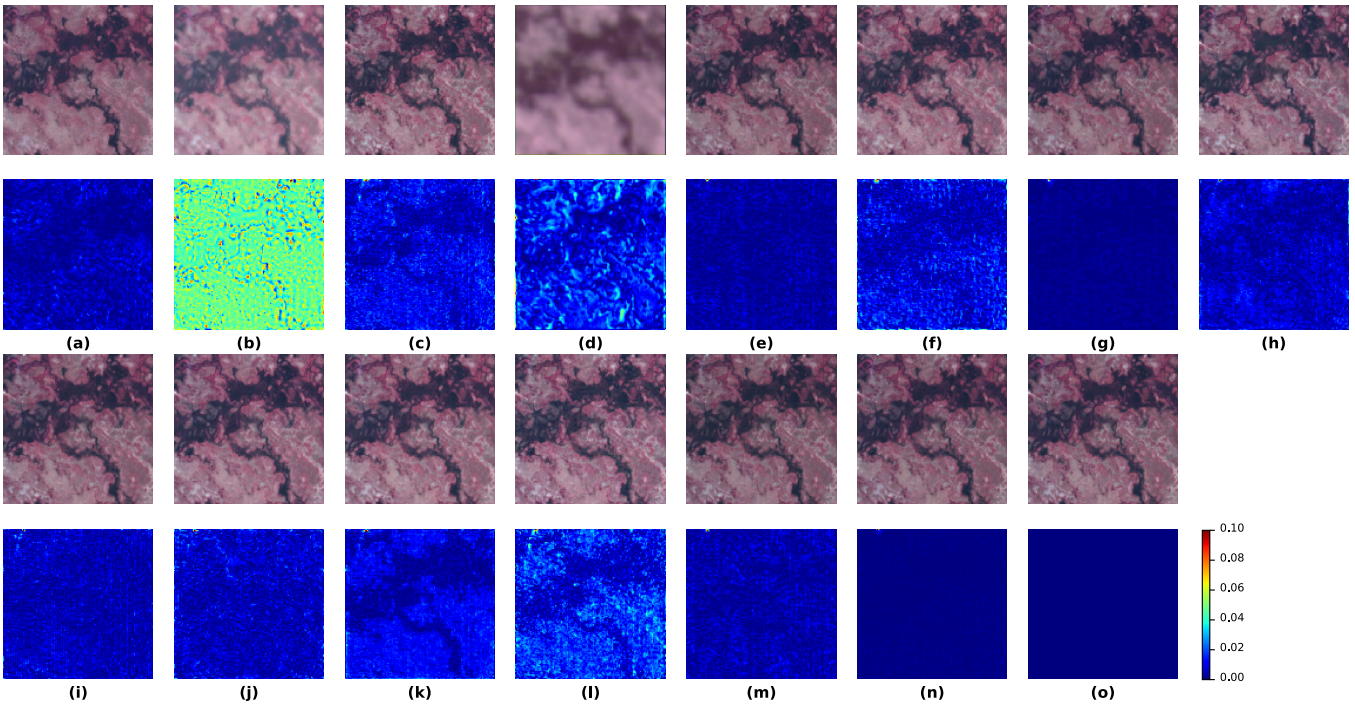


Fig. 6. Qualitative results of the Botswana dataset with pseudo-color technique. The first row presents fused images with bands 29-20-12 as RGB. The second row depicts the corresponding error images at the band of 88. (a) CNMF. (b) LTTR. (c) TFNet. (d) ResTFNet. (e) SSR-Net. (f) MoG-DCN. (g) Fusformer. (h) MCT-Net. (i) SFIN. (j) DCTransformer. (k) CYFormer. (l) FusionMamba. (m) MCIFNet. (n) CESFusion. (o) GT.

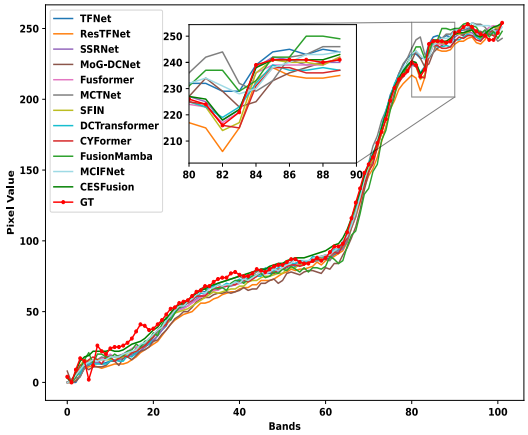


Fig. 7. Illustration of the spectral vectors of the GT and the DL-based fusion approaches at position (20, 32) in the Pavia Center dataset.

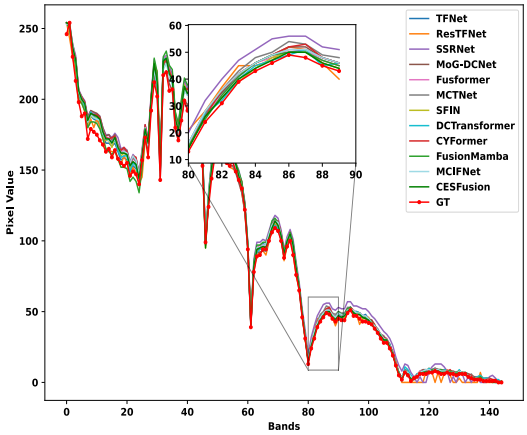


Fig. 8. Illustration of the spectral vectors of the GT and the DL-based fusion approaches at position (4, 64) in the Botswana dataset.

Furthermore, the spectral curve comparison in Fig. 7 shows pixel values of fused images generated by CESFusion are closer to the GT when compared to the other methods.

2) Results on the Botswana: In Table I, CESFusion consistently outperforms other approaches across all metrics on the Botswana dataset, showcasing its superiority in feature learning and image restoration. The PSNR and ERGAS metrics demonstrate significant improvement, underscoring the efficacy of the CIDM in effectively integrating complementary information from both HSI and MSI across spatial and frequency domains. In terms of qualitative results, as illustrated in Fig. 6, our approach exhibits appropriate detail recovery and provides minimal block artifacts. More specifically, the spectral curve comparison in Fig. 8 further demonstrates that our method produces superior fusion results compared to other methods.

3) Results on the Chikusei: As recorded in Table II, our approach achieves excellent performance across five metrics on the Chikusei dataset, corresponding to the superior ability to model frequency interactions between HSI and MSI, resulting in enhanced feature fusion. From Fig. 9, we can observe that our proposed approach produces better image quality compared to other methods, both spatially and spectrally. The error map further showcases that the spatial and spectral distortions between the output and GT are minimal.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

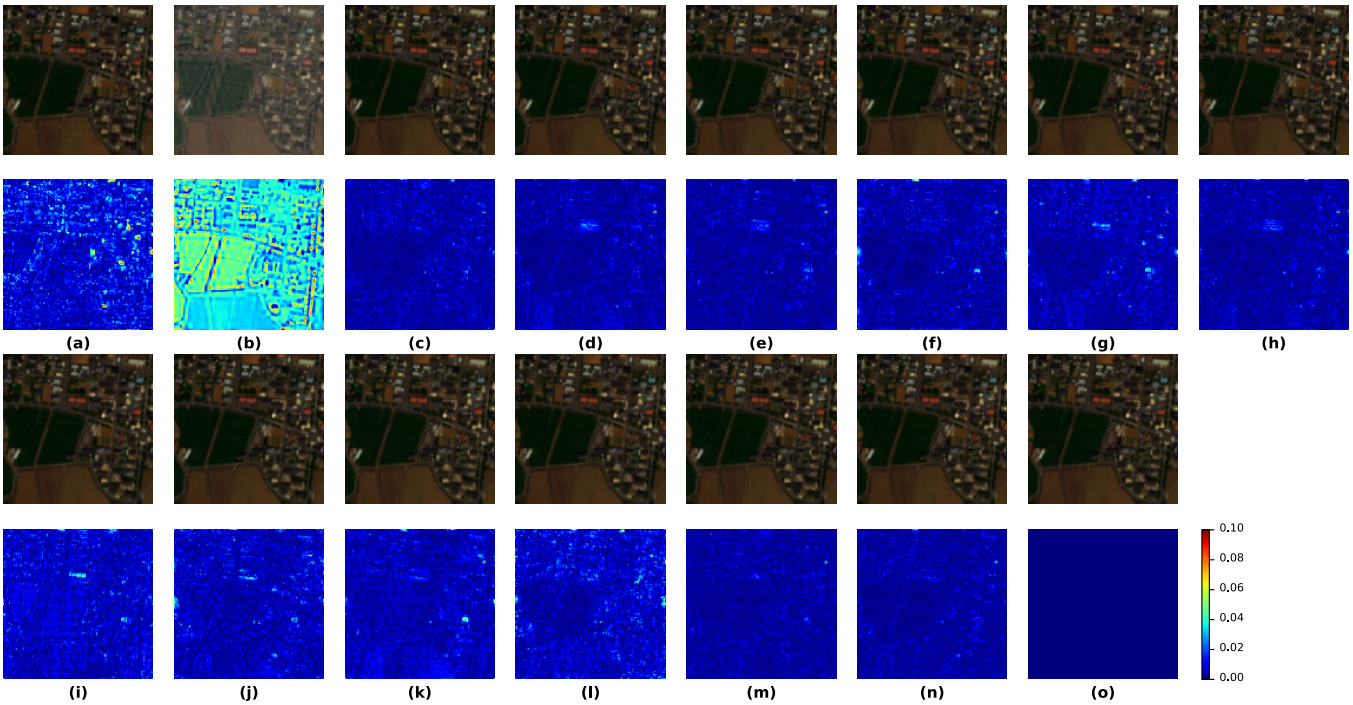


Fig. 9. Qualitative results of the Chikusei dataset with pseudo-color technique. The first row presents fused images with bands 61-35-10 as RGB. The second row depicts the corresponding error images at the band of 61. (a) CNMF. (b) LTTR. (c) TFNet. (d) ResTFNet. (e) SSR-Net. (f) MoG-DCN. (g) Fusformer. (h) MCT-Net. (i) SFIN. (j) DCTransformer. (k) CYFormer. (l) FusionMamba. (m) MCIFNet. (n) CESFusion. (o) GT.

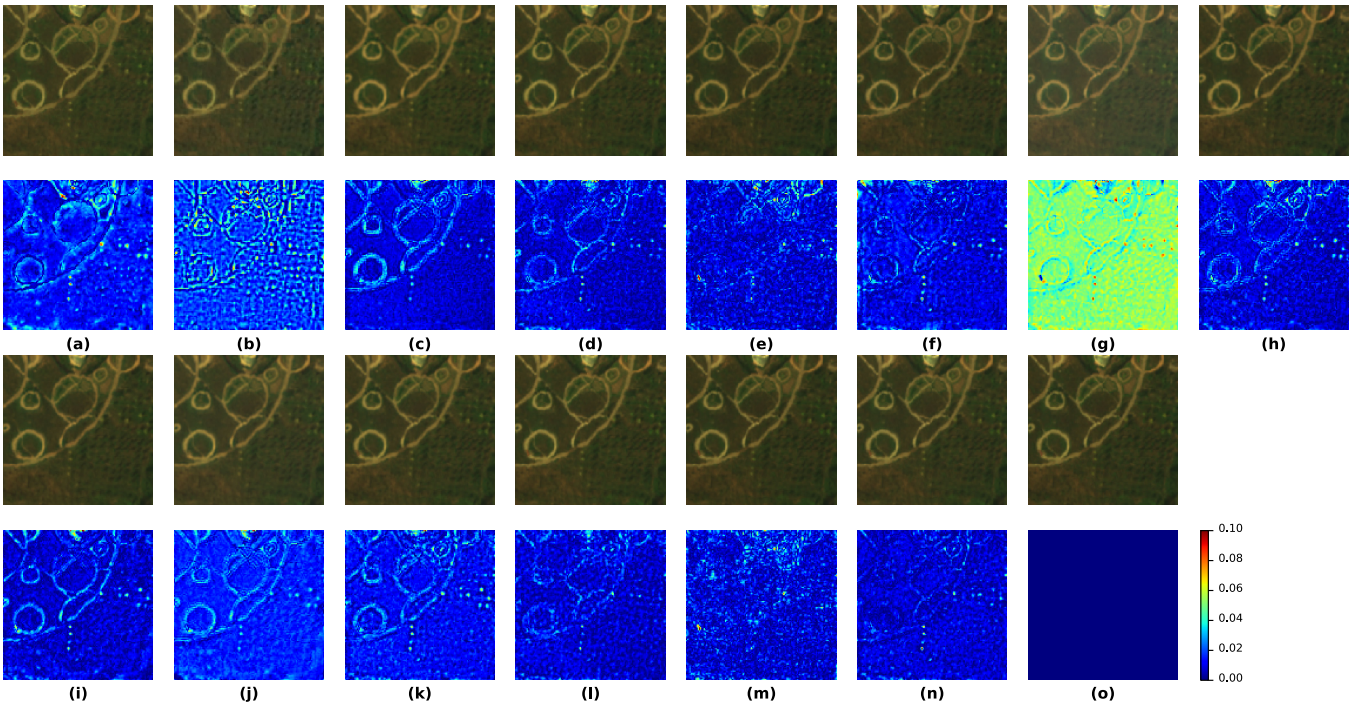


Fig. 10. Qualitative results of the XiongAn dataset with pseudo-color technique. The first row presents fused images with bans 107-62-25 as RGB. The second row depicts the corresponding error images at the band of 119. (a) CNMF. (b) LTTR. (c) TFNet. (d) ResTFNet. (e) SSR-Net. (f) MoG-DCN. (g) Fusformer. (h) MCT-Net. (i) SFIN. (j) DCTransformer. (k) CYFormer. (l) FusionMamba. (m) MCIFNet. (n) CESFusion. (o) GT.

As depicted by Fig. 11, it is evident that the spectral vectors produced by CESFusion are closer to the GT.

4) Results on the XiongAn: Table II reports the quantitative comparison results on the XiongAn dataset. Our approach outperforms other methods across five metrics, suggesting its competitive ability to integrate information from HSI and MSI, thereby achieving high-quality image fusion and restoration. MCIFNet achieves suboptimal performance evaluation, thanks to its efficient combination of the Mamba and implicit neural representation. As shown in Fig. 10, our proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

TABLE II

AVERAGE QUANTITATIVE COMPARISON OF VARIOUS METHODS ON THE CHIKUSEI AND THE XIONGAN DATASETS. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLDFACE, AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	Chikusei						XiongAn					
	PSNR↑	ERGAS↓	SAM↓	CC↑	RMSE↓	SSIM↑	PSNR↑	ERGAS↓	SAM↓	CC↑	RMSE↓	SSIM↑
CNMF [34]	31.1575	2.8530	3.9047	0.9704	0.0174	0.8612	36.7110	0.6753	0.9813	0.9667	0.0267	0.9999
LTTR [35]	29.1260	7.7258	4.2769	0.9089	0.0385	0.9089	33.8039	0.9521	1.6401	0.9368	0.0425	0.9999
TFNet [40]	46.4189	0.6082	2.7518	0.9967	0.0036	0.9967	38.9958	0.5263	0.9198	0.9768	0.0080	0.9999
ResTFNet [40]	46.7391	0.5865	2.7145	0.9969	0.0030	0.9969	42.1570	0.3520	0.6645	0.9829	0.0047	0.9999
SSR-Net [41]	45.9719	0.6151	2.7788	0.9964	0.0032	0.9963	41.9709	0.3717	0.7022	0.9826	**0.0019**	0.9999
MoG-DCN [44]	44.9141	0.6975	2.9020	0.9925	0.0030	0.9918	38.9147	0.5280	0.9898	0.9751	0.0047	0.9999
Fusformer [48]	45.3625	0.6376	2.7830	0.9958	0.0090	0.9956	41.7729	0.3764	0.6777	0.9835	0.0045	0.9999
MCT-Net [49]	45.9916	0.6006	2.7504	0.9966	0.0030	0.9966	41.2162	0.3948	0.7538	0.9813	0.0049	0.9999
SFIN [68]	45.2884	0.6070	2.7701	0.9963	**0.0023**	0.9964	41.3973	0.3978	0.7274	0.9823	0.0046	0.9999
DCTransformer [52]	42.5300	0.6302	2.8223	0.9935	0.0029	0.9930	39.0953	0.4995	0.9042	0.9728	0.0056	0.9999
CYFormer [53]	46.3201	0.5706	2.7108	0.9969	0.0029	0.9969	42.0436	0.3598	0.6775	0.9819	0.0040	0.9999
FusionMamba [62]	44.9421	0.6603	2.8330	0.9961	0.0030	0.9961	42.6166	0.3428	0.6138	0.9834	0.0069	0.9999
MCIFNet [64]	47.5429	0.5597	2.6771	0.9974	0.0029	0.9974	43.4048	0.3082	0.5726	0.9860	0.0040	0.9999
CESFusion	**48.1100**	**0.5311**	**2.6418**	**0.9975**	0.0025	**0.9975**	**43.9420**	**0.2933**	**0.5512**	**0.9868**	0.0035	**0.9999**

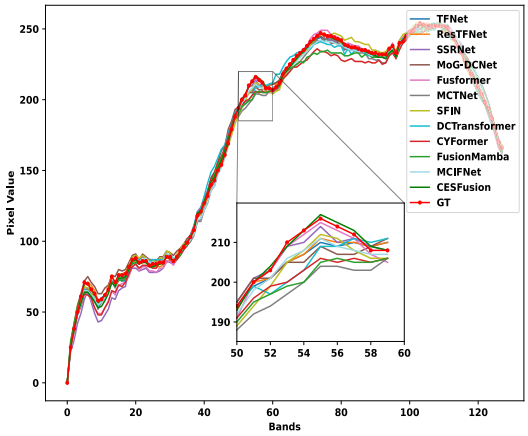


Fig. 11. Illustration of the spectral vectors of the GT and the DL-based fusion approaches at position (40, 57) in the Chikusei dataset.

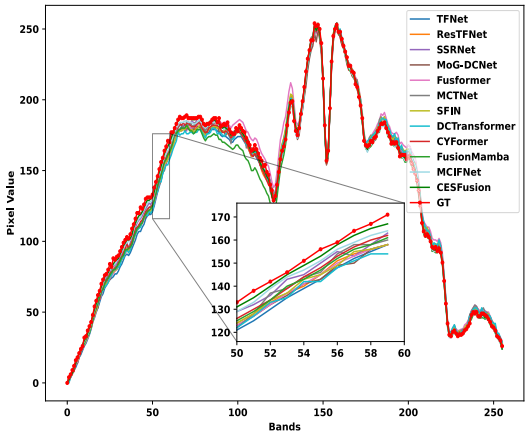


Fig. 12. Illustration of the spectral vectors of the GT and the DL-based fusion approaches at position (64, 13) in the XiongAn dataset.

CESFusion delivers fewer artifacts while faithfully preserving the spectral information from HSI, compared to other methods, demonstrating its robustness to HSI with large number of spectral bands. These can be attributed to the advantages of the proposed CESFusion, including the effective spectral feature modeling with SSM, attention to the frequency domain information, and the robust deep spatial feature learning. Furthermore, the spectral vectors depicted in Fig. 12 clearly validate the effectiveness of CESFusion in both feature extraction and image reconstruction, further narrowing the gap with the GT.

TABLE III

AVERAGE QUANTITATIVE COMPARISON OF VARIOUS METHODS ON THE WORLDVIEW DATASETS. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLDFACE, AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	$D_\lambda \downarrow$	$D_s \downarrow$	QNR ↑
TFNet [40]	0.0347	0.0391	0.9278
ResTFNet [40]	0.0602	0.0563	0.8873
SSR-Net [41]	0.0108	0.0328	0.9567
MoG-DCN [44]	0.0936	0.0791	0.8352
Fusformer [48]	0.0331	0.0365	0.9318
MCT-Net [49]	0.0135	0.0356	0.9514
SFIN [68]	0.0036	0.0340	0.9626
DCTransformer [52]	0.0031	0.0326	0.9644
CYFormer [53]	0.0039	0.0332	0.9630
FusionMamba [62]	0.0115	0.0366	0.9523
MCIFNet [64]	0.0125	0.0367	0.9512
CESFusion	**0.0023**	**0.0317**	**0.9661**

E. Experimental Results on Real-World Dataset

To further validate the efficacy of the proposed CESFusion, comprehensive comparative experiments are performed on the real-world WorldView dataset. The WorldView dataset comprises paired 3-band MSI and 8-band HSI, with a total spatial resolution of 836 × 1316 pixels. We adopt no-reference evaluation metrics, including D_λ, D_s, and QNR, to evaluate the performance of the fusion methods. Table III summarizes the average quantitative evaluation results of various approaches on the WorldView dataset. The results demonstrate that CESFusion achieves the optimal performance across all metrics, further validating its superior capability in HSI-MSI fusion. Fig. 13 presents the pseudo-color image of the fused images, providing a visual comparison of the generated images. Both ResTFNet and MoG-DCN exhibit significant spectral distortion, whereas FusionMamba demonstrates notable spatial distortion in the fused results. The proposed method achieves the optimal performance in reconstructing color and edge details, effectively enhancing spatial and spectral information, thereby demonstrating its effectiveness in real-world scenarios.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

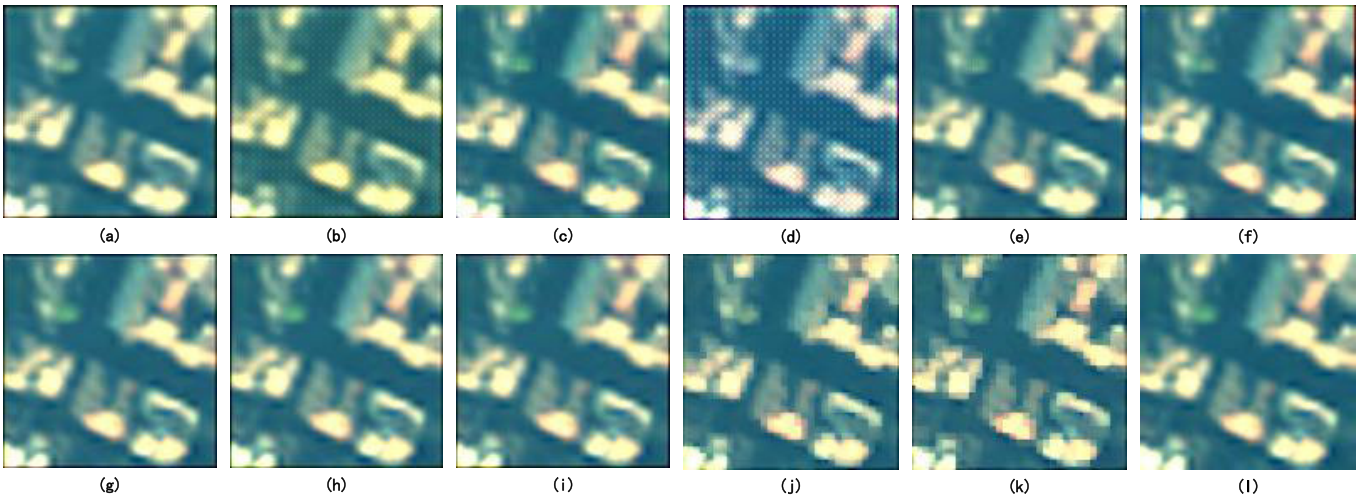


Fig. 13. Qualitative results of the real-world dataset with pseudo-color technique. (a) TFNet. (b) ResTFNet. (c) SSR-Net. (d) MoG-DCN. (e) Fusformer. (f) MCT-Net. (g) SFIN. (h) DCTransformer. (i) CYFormer. (j) FusionMamba. (k) MCIFNet. (l) CESFusion.

TABLE IV

AVERAGE QUANTITATIVE COMPARISON OF ABLATIONS ON THE PAVIA CENTER DATASET. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLDFACE

Method	PSNR	ERGAS	SAM	RMSE
CFFM + Conv	31.6277	4.3875	3.6806	0.0336
SRM + Conv	44.2567	1.4399	3.4441	0.0088
SpeMM + Conv	43.0581	1.5450	3.6806	0.0142
SRM+SpeMM+CDIM	44.1271	1.4469	3.4624	0.0097
SpeMM+CFFM+CDIM	43.8003	1.4868	3.5398	0.0011
SRM+CFFM+CDIM	44.3956	1.4249	3.4417	0.0020
SRM+SpeMM+CFFM	44.3091	1.4220	3.4664	0.0021
CESFusion	**44.4736**	**1.4176**	**3.4290**	**0.0011**

TABLE V

COMPARISON RESULTS OF DIFFERENT SPECTRAL FEATURE LEARNING MODULES ON THE PAVIA CENTER DATASET

Method	PSNR	ERGAS	SAM	RMSE
CNN-based	44.3766	1.4356	3.4297	0.0011
Transformer-based	42.5258	1.5758	3.7791	0.0039
SpeMM	**44.4736**	**1.4176**	**3.4290**	**0.0011**

TABLE VI

AVERAGE QUANTITATIVE COMPARISON UNDER DIFFERENT LOSS FUNCTION COMBINATIONS ON THE PAVIA CENTER DATASET

L_{spa}	L_{amp}	L_{pha}	PSNR	ERGAS	SAM	RMSE
✓	✗	✗	43.5469	1.5471	3.4790	0.0016
✓	✗	✓	44.0987	1.4512	3.4897	0.0013
✓	✓	✗	44.3418	1.4399	3.4920	0.0011
✓	✓	✓	**44.4736**	**1.4176**	**3.4290**	**0.0011**

F. Ablation Study

To validate the effectiveness of the main components in the proposed CESFusion, we conduct several ablation experiments on the Pavia Center dataset with a downsampling factor of 4.

1) Cross-Frequency Fusion Module Ablation Studies: We demonstrate the importance of frequency domain information in enhancing fusion performance by removing the CFFM and conducting only joint spatial–spectral feature learning. As recorded in Table IV, without the CFFM, the fusion results of our method showcase a significant decline across all evaluation metrics. This highlights the critical role of the CDIM in enhancing the model's global perception capability by effectively integrating the frequency domain information from both HSI and MSI.

2) Deep Spatial Feature Learning Module Ablation Studies: To demonstrate the necessity of the deep spatial feature learning, we remove the SRM from CESFusion and retrain the network. Without the SRM, the network can only extract spectral features from the joint spatial–spectral modeling. The quantitative results in Table IV present that spatial feature learning is essential for image fusion, enabling the recovery of high spatial fidelity in fused images.

3) SpeMM Ablation Studies: To verify the effectiveness of spectral modeling, we eliminate SpeMM while keeping the other parameters unchanged. As observed in Table IV, in the absence of SpeMM, all evaluation metrics exhibit varying degrees of degradation, indicating its effectiveness in preserving spectral continuity and improving spectral fidelity of fused images. Furthermore, to comprehensively validate the advantages of SpeMM, we conduct comparative experiments by replacing SpeMM with a CNN-based spectral feature extraction module and a Transformer-based spectral feature extraction module. As shown in Table V, the Mamba-based SpeMM achieves optimal performance across all metrics, demonstrating its effectiveness in modeling long-range spectral dependencies.

4) CDIM Ablation Studies: We replace the CDIM with a simple addition operation and a channel-wise convolutional layer to validate its necessity for fusing complementary information. The quantitative results reported in Table IV demonstrate that the critical role played by the CDIM is in integrating the cross-domain complementary information, ensuring the high-quality image fusion.

5) Investigations of Loss Function: To validate the effectiveness of loss function in improving our model's performance, we remove the L_{freq}, L_{amp}, and L_{pha}, respectively, to emphasize the importance of the frequency domain in reconstructing global information. As the quantitative results presented in Table VI, the absence of the frequency-based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

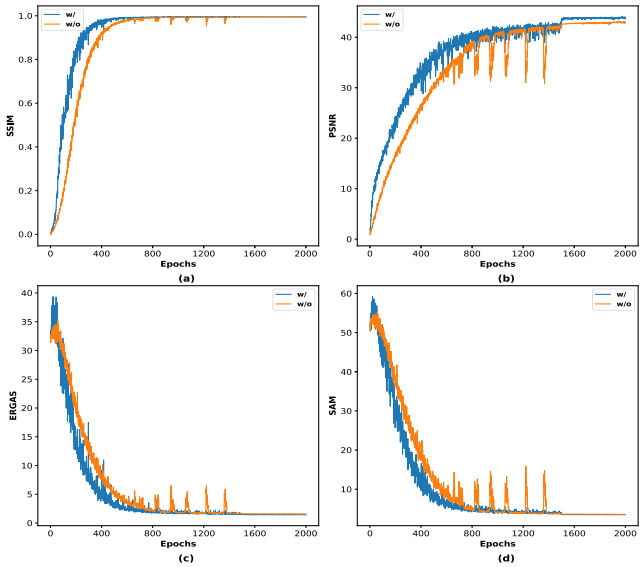


Fig. 14. Comparison of training dynamics under w/ and w/o frequency-based loss function. (a) SSIM. (b) PSNR. (c) ERGAS. (d) SAM.

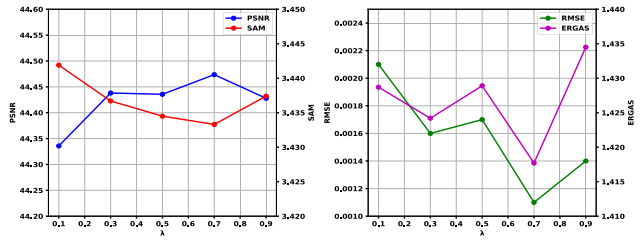


Fig. 15. Comparison of model performance under different settings of the λ in loss function.

loss constraint leads to significant performance degradation across all metrics. Fig. 14 further illustrates the dynamic changes in various metrics under constraints with and without the frequency-based loss function. These results also demonstrate that the proposed loss function effectively enhances the model's ability to reconstruct general information and improves the quality of the fused images. Moreover, we also explore the impact of different settings of λ on the fusion results. As illustrated in Fig. 15, when λ is set to 0.7, our method achieves the optimal performance on the Pavia Center dataset.

G. Analysis of Model Complexity

Table VII presents a comparison of the model parameters, computational efficiency, and inference time of our approach with other state-of-the-art methods. It can be observed that SSRNet has the fewest parameters and the smallest model, utilizing a simple CNN-based architecture for feature extraction and fusion. CYFormer boasts the fastest inference efficiency. In contrast, CESFusion has a more complicated model, larger parameter size, and longer inference time. However, despite this increased complexity, CESFusion achieves optimal fusion performance across all synthetic and real-world datasets. Incorporating frequency domain information into HSI-MSI fusion, it offers a novel approach for integrating cross-domain features, resulting in high-quality image fusion.

TABLE VII
COMPARISON RESULTS OF PARAMETERS AND FLOPS OF THE CESFUSION WITH OTHER METHODS

Method	Params (M)	FLOPs (G)	Time (ms)
TFNet [40]	2.45	9.12	722
ResTFNet [40]	2.32	8.49	235
SSR-Net [41]	**0.28**	**4.62**	173
MoG-DCN [44]	6.85	269.09	1032
Fusformer [48]	0.75	5.86	494
MCT-Net [49]	4.48	13.11	332
SFIN [68]	8.21	117.18	841
DCTransformer [52]	8.38	310.73	696
CYFormer [53]	5.86	95.80	**19**
FusionMamba [62]	4.12	22.92	32
MCIFNet [64]	6.33	59.82	100
CESFusion	5.73	39.15	348

VI. CONCLUSION

In this article, we propose a cross-frequency enhanced spatial–spectral fusion network, named CESFusion, incorporating frequency domain information into HSI-MSI fusion to enhance joint spatial–spectral feature learning. Specifically, we introduce the CFFM to integrate frequency domain information from HSI and MSI, effectively improving the model's global perceptual capability. To preserve the spectral continuity of fused HR-HSI, we propose the SpeMM to capture long-range spectral correlations based on the SSM. This is combined with the adopted SRM for deep spatial feature extraction, enabling joint spatial–spectral feature learning. The CDIM is elaborately designed to facilitate complementary interaction and fusion of cross-domain information. Compared to other state-of-the-art fusion approaches, our proposed CESFusion achieves optimal performance both quantitatively and qualitatively. However, our approach also faces challenges in parameter scalability and computational complexity. Future work will focus on lightweight strategies to enable efficient HSI-MSI fusion.

REFERENCES

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] C. Shi, S. Yue, and L. Wang, "A dual-branch multiscale transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504520.

[3] R. J. Murphy and S. T. Monteiro, "Mapping the distribution of ferric iron minerals on a vertical mine face using derivative analysis of hyperspectral imagery (430–970nm)," *ISPRS J. Photogramm. Remote Sens.*, vol. 75, pp. 29–39, Jan. 2013.

[4] C. Wang et al., "A review of deep learning used in the hyperspectral image analysis for agriculture," *Artif. Intell. Rev.*, vol. 54, no. 7, pp. 5205–5253, Oct. 2021.

[5] R. Sarić et al., "Applications of hyperspectral imaging in plant phenotyping," *Trends Plant Sci.*, vol. 27, no. 3, pp. 301–315, Mar. 2022.

[6] A. Baiano, C. Terracone, G. Peri, and R. Romaniello, "Application of hyperspectral imaging for prediction of physico-chemical and sensory characteristics of table grapes," *Comput. Electron. Agricult.*, vol. 87, pp. 142–151, Sep. 2012.

[7] B. Huang, H. Song, H. Cui, J. Peng, and Z. Xu, "Spatial and spectral image fusion using sparse matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 3, pp. 1693–1704, Mar. 2014.

[8] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

[9] M. A. Veganzones, M. Simoes, G. Licciardi, N. Yokoya, J. M. Bioucas-Dias, and J. Chanussot, "Hyperspectral super-resolution of locally low rank images from complementary multisource data," *IEEE Trans. Image Process.*, vol. 25, no. 1, pp. 274–288, Jan. 2016.

[10] L. Sun and H. Guo, "Blind unmixing of hyperspectral images based on L_1 norm and tucker tensor decomposition," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[11] Q. Zhang, Y. Dong, Q. Yuan, M. Song, and H. Yu, "Combined deep priors with low-rank tensor factorization for hyperspectral image restoration," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[12] Y. Chen, J. Zeng, W. He, X.-L. Zhao, and T.-Z. Huang, "Hyperspectral and multispectral image fusion using factor smoothed tensor ring decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515417.

[13] L. Alparone, S. Baronti, B. Aiazzi, and A. Garzelli, "Spatial methods for multispectral pansharpening: Multiresolution analysis demystified," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2563–2576, May 2016.

[14] S. Zhong, Y. Zhang, Y. Chen, and D. Wu, "Combining component substitution and multiresolution analysis: A novel generalized BDSD pansharpening algorithm," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 2867–2875, Jun. 2017.

[15] X. Fu, Z. Lin, Y. Huang, and X. Ding, "A variational pan-sharpening with local gradient constraints," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10257–10266.

[16] G. Vivone, R. Restaino, and J. Chanussot, "Full scale regression-based injection coefficients for panchromatic sharpening," *IEEE Trans. Image Process.*, vol. 27, no. 7, pp. 3418–3431, Jul. 2018.

[17] Q. Xu, B. Li, Y. Zhang, and L. Ding, "High-fidelity component substitution pansharpening by the fitting of substitution data," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7380–7392, Nov. 2014.

[18] L. Deng, G. Vivone, W. Guo, M. D. Mura, and J. Chanussot, "A variational pansharpening approach based on reproducible kernel Hilbert space and heaviside function," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4330–4344, Sep. 2018.

[19] R. Dian, S. Li, B. Sun, and A. Guo, "Recent advances and new guidelines on hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 69, pp. 40–51, May 2021.

[20] K. Li, W. Zhang, D. Yu, and X. Tian, "HyperNet: A deep network for hyperspectral, multispectral, and panchromatic image fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 188, pp. 30–44, Jun. 2022.

[21] Z. Wang, B. Chen, R. Lu, H. Zhang, H. Liu, and P. K. Varshney, "FusionNet: An unsupervised convolutional variational network for hyperspectral and multispectral image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 7565–7577, 2020.

[22] R. Ran, L.-J. Deng, T.-J. Zhang, J. Chang, X. Wu, and Q. Tian, "KNLConv: Kernel-space non-local convolution for hyperspectral image super-resolution," *IEEE Trans. Multimedia*, vol. 26, pp. 8836–8848, 2024.

[23] S. Liu, S. Liu, S. Zhang, B. Li, W. Hu, and Y.-D. Zhang, "SSAU-net: A spectral–spatial attention-based U-net for hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542116.

[24] S. Xu, O. Amira, J. Liu, C.-X. Zhang, J. Zhang, and G. Li, "HAM-MFN: Hyperspectral and multispectral image multiscale fusion network with RAP loss," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4618–4628, Jul. 2020.

[25] T. You, C. Wu, Y. Bai, D. Wang, H. Ge, and Y. Li, "HMF-Former: Spatio-spectral transformer for hyperspectral and multispectral image fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[26] L. Chen, G. Vivone, J. Qin, J. Chanussot, and X. Yang, "Spectral–spatial transformer for hyperspectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 16733–16747, Nov. 2024.

[27] J. Sun, B. Chen, R. Lu, Z. Cheng, C. Qu, and X. Yuan, "Advancing hyperspectral and multispectral image fusion: An information-aware transformer-based unfolding network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 7407–7421, Apr. 2025.

[28] L. Sun et al., "MDC-FusFormer: Multiscale deep cross-fusion transformer network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528914.

[29] S. Jia, Z. Min, and X. Fu, "Multiscale spatial–spectral transformer network for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 96, pp. 117–129, Aug. 2023.

[30] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[31] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-mamba: 3D-spectral–spatial mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.

[32] W. Zhou, S.-I. Kamata, H. Wang, M. S. Wong, and H. Hou, "Mamba-in-mamba: Centralized mamba-cross-scan in tokenized mamba model for hyperspectral image classification," *Neurocomputing*, vol. 613, Jan. 2025, Art. no. 128751.

[33] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 3586–3594.

[34] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[35] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.

[36] J. Xue, Y.-Q. Zhao, Y. Bu, W. Liao, J. C.-W. Chan, and W. Philips, "Spatial–spectral structured sparse low-rank representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 3084–3097, 2021.

[37] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Nonlocal patch tensor sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 3034–3047, Jun. 2019.

[38] J. Xue, Y.-Q. Zhao, T. Wu, and J. C.-W. Chan, "Tensor convolution-like low-rank dictionary for high-dimensional image representation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 13257–13270, Dec. 2024.

[39] G. Vivone et al., "Deep learning in remote sensing image fusion: Methods, protocols, data, and future perspectives," *IEEE Geosci. Remote Sens. Mag. Replaces Newslett.*, vol. 13, no. 1, pp. 269–310, Mar. 2025.

[40] X. Liu, Q. Liu, and Y. Wang, "Remote sensing image fusion based on two-stream fusion network," *Inf. Fusion*, vol. 55, pp. 1–15, Mar. 2020.

[41] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.

[42] Q. Xie, M. Zhou, Q. Zhao, D. Meng, W. Zuo, and Z. Xu, "Multispectral and hyperspectral image fusion by MS/HS fusion net," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1585–1594.

[43] R. Ran, L.-J. Deng, T.-X. Jiang, J.-F. Hu, J. Chanussot, and G. Vivone, "GuidedNet: A general CNN fusion framework via high-resolution guidance for hyperspectral image super-resolution," *IEEE Trans. Cybern.*, vol. 53, no. 7, pp. 4148–4161, Jul. 2023.

[44] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[45] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[46] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[47] D. Liu et al., "An efficient unfolding network with disentangled spatial–spectral representation for hyperspectral image super-resolution," *Inf. Fusion*, vol. 94, pp. 92–111, Jun. 2023.

[48] J.-F. Hu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, D. Hong, and G. Vivone, "Fusformer: A transformer-based fusion network for hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] X. Wang, X. Wang, R. Song, X. Zhao, and K. Zhao, "MCT-net: Multi-hierarchical cross transformer for hyperspectral and multispectral image fusion," *Knowledge-Based Syst.*, vol. 264, Mar. 2023, Art. no. 110362.

[50] Y. Xiao, Q. Yuan, K. Jiang, J. He, C.-W. Lin, and L. Zhang, "TTST: A top-k token selective transformer for remote sensing image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 738–752, 2024.

[51] S. Deng, L.-J. Deng, X. Wu, R. Ran, and R. Wen, "Bidirectional dilation transformer for multispectral and hyperspectral image fusion," in *Proc. 32nd Int. Joint Conf. Artif. Intell.*, Aug. 2023, pp. 3633–3641.

[52] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Reciprocal transformer for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102148.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.

[53] S. Chen, L. Zhang, and L. Zhang, "Cyclic cross-modality interaction for hyperspectral and multispectral image fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 1, pp. 741–753, Jan. 2025.
[54] J. Fang, J. Yang, A. Khader, and L. Xiao, "Deep unfolding network enhanced by transformer priors for unregistered hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5531616.
[55] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.
[56] J. T. H. Smith, A. Warrington, and S. W. Linderman, "Simplified state space layers for sequence modeling," 2022, *arXiv:2208.04933*.
[57] D. Liao, Q. Wang, T. Lai, and H. Huang, "Joint classification of hyperspectral and LiDAR data based on mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530915.
[58] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-mamba for large remote sensing image dense prediction," 2024, *arXiv:2404.02668*.
[59] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.
[60] G. Zhao, H. Wu, D. Luo, X. Ou, and Y. Zhang, "Spatial–spectral interaction super-resolution CNN–mamba network for fusion of satellite hyperspectral and multispectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 18489–18501, 2024.
[61] H. Zhou, X. Wu, H. Chen, X. Chen, and X. He, "RSDehamba: Lightweight vision mamba for remote sensing satellite image dehazing," 2024, *arXiv:2405.10030*.
[62] S. Peng, X. Zhu, H. Deng, L.-J. Deng, and Z. Lei, "FusionMamba: Efficient remote sensing image fusion with state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5410216.
[63] Y. Xiao, Q. Yuan, K. Jiang, Y. Chen, Q. Zhang, and C.-W. Lin, "Frequency-assisted mamba for remote sensing image super-resolution," *IEEE Trans. Multimedia*, vol. 27, pp. 1783–1796, 2025.
[64] C. Zhu, S. Deng, X. Song, Y. Li, and Q. Wang, "Mamba collaborative implicit neural representation for hyperspectral and multispectral remote sensing image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504915.
[65] M. Zhou et al., "Adaptively learning low-high frequency information integration for pan-sharpening," in *Proc. 30th ACM Int. Conf. Multimedia*, New York, NY, USA, Oct. 2022, pp. 3375–3384.
[66] Y. Yang and S. Soatto, "FDA: Fourier domain adaptation for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4084–4094.
[67] Z. Huang, Z. Zhang, C. Lan, Z.-J. Zha, Y. Lu, and B. Guo, "Adaptive frequency filters as efficient global token mixers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6026–6036.
[68] M. Zhou et al., "A general spatial-frequency learning framework for multimodal image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, pp. 1–18, Feb. 2024.
[69] N. Zheng, M. Zhou, J. Huang, and F. Zhao, "Frequency integration and spatial compensation network for infrared and visible image fusion," *Inf. Fusion*, vol. 109, Sep. 2024, Art. no. 102359. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253524001374
[70] G. Yang, J. Li, and X. Gao, "A dual domain multi-exposure image fusion network based on spatial-frequency integration," *Neurocomputing*, vol. 598, Sep. 2024, Art. no. 128146.
[71] F. Dell'Acqua, P. Gamba, A. Ferrari, J. A. Palmason, J. A. Benediktsson, and K. Árnason, "Exploiting spectral and spatial information in hyperspectral urban data with high resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 4, pp. 322–326, Oct. 2004.
[72] S. G. Ungar, J. S. Pearlman, J. A. Mendenhall, and D. Reuter, "Overview of the Earth observing one (EO-1) mission," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1149–1159, Jun. 2003.
[73] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Dept. Space Appl. Lab., Univ. Tokyo, Bunkyō, Japan, Tech. Rep. SAL-2016-05-27, 2016, vol. 5, no. 5, p. 5.
[74] Y. Cen et al., "Aerial hyperspectral remote sensing classification dataset of Xiongan new area (Matiwan village)," *J. Remote Sens.*, vol. 24, no. 11, pp. 1299–1306, 2020.
[75] T. Ranchin and L. Wald, "Fusion of high spatial and spectral resolution images: The ARSIS concept and its implementation," *Photogramm. Eng. Remote Sens.*, vol. 66, no. 1, pp. 49–61, Jan. 2000.



Haozheng Zhang is currently pursuing the Ph.D. degree with the Key Laboratory of Computer Vision and System, Ministry of Education, Tianjin University of Technology, Tianjin, China.

His research interests include computer vision and image analysis.



Yanhong Yang received the Ph.D. degree in control science and engineering from Zhejiang University of Technology, Hangzhou, China, in 2020.

She is currently a Lecturer with Tianjin University of Technology, Tianjin, China. Her research interests include image processing and low-rank representation.



Chaoyang Li received the B.S. degree in computer science and technology from Ningbo University of Finance and Economics, Ningbo, China, in 2023. He is currently pursuing the master's degree with the Key Laboratory of Computer Vision and System, Ministry of Education, Tianjin University of Technology, Tianjin, China.

His research interests include computer vision and defect detection.



Yanjie Lu is currently pursuing the Ph.D. degree with the Institute of Intelligent Media Computing, Hangzhou Dianzi University, Hangzhou, China.

His research interests include computer vision and image analysis.



Guodao Zhang (Member, IEEE) received the Ph.D. degree from Zhejiang University of Technology, Hangzhou, China, in 2022.

His main research interests include machine learning and computational intelligence.



Shengyong Chen (Senior Member, IEEE) received the Ph.D. degree from the City University of Hong Kong, Hong Kong, in 2003.

He worked as a Guest Researcher at the University of Hamburg, Hamburg, Germany, where he received a fellowship from the Alexander von Humboldt Foundation in 2006. He was a Visiting Professor at the Imperial College London, London, U.K., from 2008 to 2009. He is currently a Full Professor with Tianjin University of Technology. He has over 100 patents and has published over 400 scientific articles. His research interests include computer vision and robotics.

Dr. Chen is a fellow of IET.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:55:56 UTC from IEEE Xplore. Restrictions apply.